

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Safety, Immunogenicity, and Efficacy of V181 Dengue Vaccine in Healthy Participants 2 to 17 Years of Age **Short Title/Acronym:** MOBILIZE-1 / V181-005 **Protocol Number:** V181-005 **NCT Number:** NCT07013487 **Sponsor Name:** Merck Sharp & Dohme (MSD) **Investigational Product:** V181 (Live attenuated quadrivalent dengue vaccine) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Clinical Operations Lead [Contact Information on File]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of a single dose of V181 in preventing symptomatic virologically confirmed dengue (VCD) of any severity, due to any of the four dengue serotypes (DENV-1, DENV-2, DENV-3, and DENV-4), regardless of prior dengue exposure. **Secondary Objectives:** To evaluate the reactogenicity and long-term immunogenicity (Dengue virus-neutralizing antibody titers) of V181.

2.2 Background & Rationale Dengue disease poses a critical global public health challenge, with approximately half of the world's population (four billion people) living in areas at risk for infection. Around 105 million dengue viral infections occur annually, leading to severe illness and an average of 29,000 deaths worldwide each year. The MOBILIZE-1 study addresses an unmet medical need by evaluating V181 as a single-dose vaccination option designed to protect against all four dengue serotypes, regardless of a patient's prior infection status.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 12,000 healthy individuals **Number of Sites:** >30 trial sites across dengue-endemic areas in the Asia-Pacific region (e.g., Indonesia, Malaysia, Philippines, Singapore, Thailand, and Vietnam). **Estimated Study Start:** 11-Jun-2025 **Estimated Primary Completion:** 24-Oct-2031

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy individuals. 2. Age ≥ 2 to ≤ 17 years of age. 3. Ability of parent/guardian to understand and provide informed consent.

4.2 Exclusion Criteria 1. Known or suspected impairment of immunological function, or history of congenital or acquired immunodeficiency. 2. Documented HIV, Hepatitis B, or Hepatitis C infection. 3. Previous receipt of a dengue vaccine or plans to receive any dengue vaccine (investigational or approved) during the study. 4. Bleeding disorder contraindicating subcutaneous vaccination or repeated venipuncture.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of V181 or matching Placebo. **Route of Administration:** Subcutaneous. **Duration of Treatment:** Single administration with short-term reactogenicity follow-up (28 days) and long-term efficacy/immunogenicity follow-up (up to 5 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard medications for routine childhood illnesses that do not interfere with immune response. **Prohibited:** Concurrent administration of other investigational drugs or any other dengue vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening/Baseline	Day 0	Informed Consent, Medical History, Baseline Blood Draw (Serostatus), Randomization, Single Dose Vaccination.
Interim 1	Day 28	Reactogenicity and Immunogenicity assessment for subset participants.
Fever Surveillance	Continuous	Monitoring for body temperature $\geq 100.4^{\circ}\text{F}$ and subsequent virological testing.
End of Study	Up to Year 5	Long-term immunogenicity (VRNT) subset evaluation and final efficacy analyses.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Frequency of symptomatic virologically confirmed dengue (VCD) of any severity, due to any of the 4 dengue serotypes. **Measurement Method:** Dengue status confirmed by WT RT-PCR or NS1 ELISA.

7.2 Secondary & Exploratory Endpoints * Dengue virus-neutralizing antibody titers (DENV-1, DENV-2, DENV-3, DENV-4) measured by Virus Reduction Neutralization Test (VRNT) up to 5 years

postvaccination. * Reactogenicity and safety events up to 28 days postvaccination.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of reactogenicity events (e.g., injection site reactions, fever) via patient/parent diaries up to 28 days postvaccination. **Serious Adverse Events (SAEs):** Actively monitored and reported continuously throughout the trial duration. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to review safety data on an ongoing basis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for reactogenicity and overall safety. **Sample Size Justification:** $N=12,000$ to provide adequate statistical power to demonstrate efficacy of V181 against VCD caused by any of the four dengue serotypes compared to placebo. **Handling of Missing Data:** Standard imputation methods for vaccine efficacy trials as prespecified in the SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence, subject safety, and data integrity. **Audit Trail:** Continuous eCRF locking and data clarification workflows.

11. Study Identifiers & Governance

Primary ID: V181-005 **Registry Number:** NCT07013487 **Sponsor/Lead Organization:** Merck Sharp & Dohme (MSD) **Study Title:** MOBILIZE-1 **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Safety, Immunogenicity, and Efficacy of V181 Dengue Vaccine in Healthy Participants 2 to 17 Years of Age

12. Clinical Framework (Dengue Specific)

Primary Condition: Dengue Disease **Diagnosis Threshold:** Fever $\geq 100.4^{\circ}\text{F}$ confirmed by WT RT-PCR or NS1 ELISA **Medical Methodology:** Prophylactic Vaccination (Live Attenuated Quadrivalent Vaccine) **Optimization Target:** Prevention of

symptomatic virologically confirmed dengue (VCD) and long-term neutralizing antibody seropositivity

13. Study Procedures & Methodology

Management Pathway: Active surveillance for febrile illness and prompt virological testing. **Education Protocol (HCP):** Training on protocol procedures, blood sample processing (RT-PCR/ELISA), and fever evaluation. **Education Protocol (Patient):** Caregiver education on temperature monitoring, fever reporting, and completion of reactogenicity eDiaries. **Quality Audit Mechanism:** Source data verification and centralized laboratory monitoring for immunogenicity subsets.

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years (long-term immunogenicity subset)
Onsite Visit Cadence: Baseline, Day 28 (for subset), and scheduled follow-ups for suspected dengue evaluation. **Remote Monitoring Frequency:** Regular fever surveillance contacts.
Checklist Review Interval: During scheduled follow-up visits.

15. Sign-off & Approval

Role Name	Signature	Date	:--	:--	:--	
Principal Investigator	_____	[DD-MMM-YYYY]			Clinical	
Operations Lead	_____	[DD-MMM-YYYY]			Lead Statistician	
	_____	[DD-MMM-YYYY]				